

# ***TMS320DM270***

## ***CPU Reference Guide***

# ***DM270 vs. DSC25***

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| Revision | Date        | Notes                                           |
|----------|-------------|-------------------------------------------------|
| 0.1      | 28 Apr 2003 | (MM) Initial Draft                              |
| 0.2      | 4 May 2003  | (SR) Put in standard format, add summary tables |

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# 1 Introduction

This document describes the differences between the TMS320DM270 (DM270) and TMS320DSC25 (DSC25) Digital Media Processors. Prior to publishing a full DM270 Technical Reference Manual (TRM), this document should be used in conjunction with the *TMS320DSC25 Technical Reference Manual* as a programming/users guide for the DM270. Please refer to the *TMS320DM270 ARM Register Manual* and *TMS320DM270 DSP Register Manual* when registers are referenced as details and memory maps will be different in many cases.

## 1.1 New Features in DM270

*Table 1-1 New Features in DM270 (vs. DSC25)*

| <b>New Features</b>                      | <b>Description</b>                                                                                                                                                                                                                             |
|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hardware 3A                              | Hardware module to compute statistics for AE, AWB, AF                                                                                                                                                                                          |
| I2C Interface                            | 1 channel I2C port, single bus master                                                                                                                                                                                                          |
| ARM7 Instruction cache                   | 8KB (Cache External Memory and SDRAM)                                                                                                                                                                                                          |
| ARM Address Trace                        | Traces ARM program execution                                                                                                                                                                                                                   |
| ARM Boot Modes                           | New Boot Modes:<br>- Self Boot (from flash) as in DCS25<br>- Boot from Serial port<br>- Boot from External CPU                                                                                                                                 |
| Direct DMA access to EMIF/SDRAM from DSP | HPIB is connected to SDRAM/Memory Traffic Controller to provide the DSP a new DMA channel to EMIF/SDRAM. (In DSC25, only have Image Buffers $\leftrightarrow$ SDRAM/EMIF DMA path and data then had to be transferred to DSP internal memory.) |

## 1.2 Modified Modules in DM270

*Table 1-2 Modified Modules in DM270 (vs. DSC25)*

| <b>Module</b>                            | <b>Addition/Modification</b>                                                                                                                                                                                                                                                                                                                                            | <b>Deletion</b>                                                                                                                                                            |
|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Timers                                   | No additions or modifications                                                                                                                                                                                                                                                                                                                                           | Removed TMCLKx registers because the bits to select between the reference clock (27MHz) and the PLL generated clock (ARM_CLK?) are now in the CLKC module: CLKC, bits 8:5. |
| Serial Interface                         | Added:<br>- SIF circuit clear capability SIOMODE:SIOCLR<br>- Master/Slave selection, SIOMODE:MSSEL                                                                                                                                                                                                                                                                      | Removed:<br>- DMA transfer forced termination bit, DMATRG0:BRK<br>- DMA permission bit, DMASTAT0:DMAP                                                                      |
| UART                                     | No change                                                                                                                                                                                                                                                                                                                                                               | No change                                                                                                                                                                  |
| Watch Dog Timer                          | - Added bits to WDTMD for testing (bits 7:3 and 1).<br>- External host signal, WDTXRST register<br>Need clarification:<br>- Modification of output additional register address outside 0<br>- Additional reset signals                                                                                                                                                  |                                                                                                                                                                            |
| MMC/SD                                   | - Timeout for DMA, DMATOR<br>- Added registers specific to SDIO mode – 4 new registers – SDIOCTL, SDIOST0, SDIOIEN, SDIOIST                                                                                                                                                                                                                                             |                                                                                                                                                                            |
| Interrupt Controller                     | Additional interrupts sources, from 28 to 39, including 8 additional external interrupts from GIO                                                                                                                                                                                                                                                                       |                                                                                                                                                                            |
| GIO                                      | - Added selection of rising, falling, or either edge for interrupt trigger.<br>- Added registers to ignore chatter on GIO signals (debouncing?)                                                                                                                                                                                                                         |                                                                                                                                                                            |
| DSPC                                     | 64 Kword<br>RAM + DMA path via HPIB                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                            |
| OSD                                      |                                                                                                                                                                                                                                                                                                                                                                         | Removed bitmap cursor window and associated registers/fields                                                                                                               |
| CCD Controller                           | - DCSUB – more bits<br>- MEDFILT – additional threshold bits, enable bit moved<br>- Functional change to Gamma adjustment and associated registers<br>- MODESET – reflects functional changes and new CCDCFG register with additional mode/config settings<br>- Changes to HSIZE register and new SDOFST register<br>- Additional input/output data patterns and timing | Removed user configurable gamma table and associated registers                                                                                                             |
| Preview Engine                           | - PVSET1 – added 8-bit/10-bit selection and inverse A-Law modes<br>- Functional change to White Balance and associated registers, now single gain values for each color sample, WBGAIN0, WBGAIN1                                                                                                                                                                        | Deleted piecewise linear White Balance adjustment and associated registers, WBGAINx, WBBPx, WBOFSTx                                                                        |
| Video Encoder and Digital RGB/YUV output | - Supports only composite output (single DAC)<br>- Added DAC powerdown capability<br>- Added capability to specify separate DCLK masks for even/odd lines<br>- GCPDATA – Added ability to write/read GCP data<br>- EPSON_LCD – control support of Epson LCD<br>- CASIO – control support of Casio LCD<br>- DOUTCTL – additional digital output controls                 | Deleted 2 DACs                                                                                                                                                             |

| <b>Module</b>             | <b>Addition/Modification</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Deletion</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Clock Controller          | <ul style="list-style-type: none"> <li>- Changed from 3 PLLs (ARM, DSP, SDRAM) to 2</li> <li>- Added SEL register to select clock source (PLLA/PLLB) for ARM, DSP, SDRAM, and Accelerators.</li> <li>- More module clock selections can be made in CLKC</li> <li>- DIV – settings apply to module clock, not PLL</li> <li>- BYP – changes to bypass PLL for each module clock</li> <li>- Addition of 3 output clock to GIO, with selectable frequency division, OxDIV</li> <li>- Addition of two GIO PWM outputs, PWMxC/PWMxH</li> </ul>                                                                             | <ul style="list-style-type: none"> <li>- Removed PLLSETx registers and capabilities (charge pump control, etc)</li> <li>- Removed SDRAM PLL</li> <li>- Removed MMC clock inversion capability, MMCLK, bit 8</li> <li>- Removed MSCLK (clock dividide ratio for MS)</li> </ul>                                                                                                                                                                                           |
| Bus Controller            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Moved EMSETx/EMSZx registers for external memory configuration to EMIF module.                                                                                                                                                                                                                                                                                                                                                                                          |
| SDRAM Controller          | <ul style="list-style-type: none"> <li>- SDBUFCTL – additional capability to disable SDRAM access by Preview, 3A, and Ext. CPU and allow access of the FIFO. (I think this means that if a module is selected, the contents of its SDRAM burst FIFO can be read from the SDBUFxx registers in the SDRAM controller.</li> <li>- SDMODE – added settting for tcRCD</li> <li>- REFCTL – added selection for DSP (HPIB) DMA access → width/position of SDMAx fields changed.</li> <li>- Added SDPTY10 and modified meaning for other SDPTY registers to support Hardware3A and Ext Host CPU SDRAM DMA access.</li> </ul> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| External Memory Interface | <ul style="list-style-type: none"> <li>- Added ability to share SDRAM bus with CS0-4, setting in BUSRLS:BUSSEL.</li> <li>- (ExtCpu mode change functional modification) ???</li> <li>- PRIORCTL – changed settings and default values</li> <li>- DPSTRx – new registers (from BUSC) but only set largest address for each region, rather that starting address and size</li> <li>- TEST – new register, can output ARM internal addresses when external memory is idle.</li> </ul>                                                                                                                                   | <ul style="list-style-type: none"> <li>- Removed ability to used upper 16-bits of SDRAM bus for memory card busses when in 16-bit SDRAM mode, BUSCTRL field deletions (see changes in BUSRLS, this may not have been deleted completely).</li> <li>- BUSINTEN – removed all interrupt conditions except CF ready/busy. Also affects BUSINTSTS, which is deleted because there is only 1 interrupt cause and BUSSTS to remove the STSCHG and CD1/2 conditions</li> </ul> |
| USB                       | <ul style="list-style-type: none"> <li>- Added support for Endpoint 4 (input) and associated (EP4lxx) registers</li> <li>- Changed register addresses</li> <li>- DMARES – new, shows number of bytes remaining in DMA transfer</li> <li>- DMAST – new, DMA status</li> </ul>                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Memory Stick              | Substantial changes – see additional material under license                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| External Host CPU         | Substantial changes – see additional material                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| DSP Accelerators          | iMX: enhanced clock speed (~2x DSP clock)<br>VLCD: added MPEG-2 support                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

## 2 Preview Engine

### 2.1 White Balance:

The DSC25's piecewise linear White Balance adjustment and associated registers for each of 3 colors has been changed to a single gain value for each of 4 color elements.

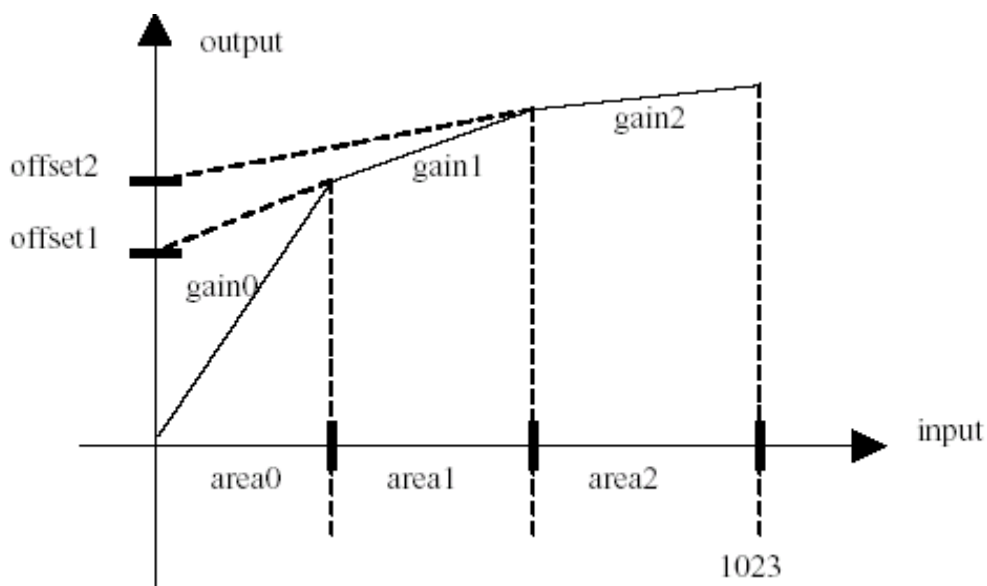


Figure 2-1 DSC25 white balance adjuster

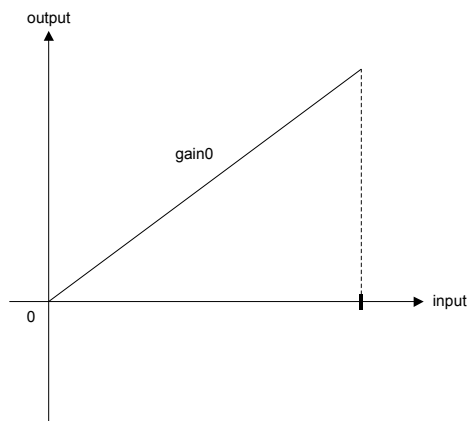


Figure 2-2 DM270 white balance adjuster



## 2.2 Select Input Mode 2

The preview engine in DM270 has three selectable input modes in SDRAM input mode.

- 10-bit mode (PVET1:8BITS = 0 and PVSET1:ALAW = 0 and PVSET1:INMOD = 1)
- 8-bit mode (PVSET1:8BITS = 1 and PVSET1:ALAW = 0 and PVSET1:INMOD = 1)
- Inverse A-Law mode (PVSET1:8BITS = 1 and PVSET1:ALAW = 1 and PVSET1:INMOD = 1)

DSC25 has only 10-bit mode.

### 2.2.1 10-bit mode

Same as DSC25.

### 2.2.2 8-bit

In this mode, the data processed from CCD Controller or SDRAM is assumed to be in 8-bit packed format. This allows the Preview Engine to postprocess data that was stored to SDRAM in the CCD Controller's 8-bit packed mode.

Note that the data is expanded to the normal 10-bit resolution before processing.

### 2.2.3 Inverse A-Law Mode

This mode allows the Preview Engine to postprocess data that was stored to SDRAM with the CCD Controller's A-Law compression applied. An inverse A-Law conversion is applied prior to Preview Engine processing.

## 3 On Screen Display

### 3.1 Bitmap Cursor

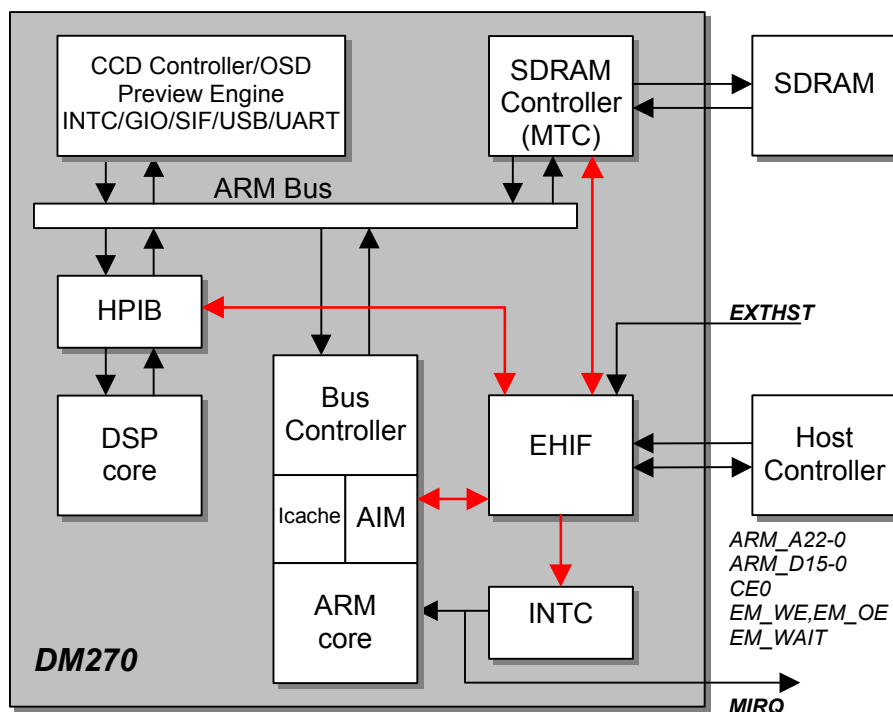
The bitmap cursor displays bitmap data stored in the OSD's internal RAM. This feature is available in DSC25 but not in DM270.

## 4 External Host Interface

This module allows a host device to access AIM, DSP Internal memory and SDRAM connected to DM270. DM270 EHIF has new features compared to DSC25.

### 4.1 FUNCTION

#### 4.1.1 Block Diagram



By setting the pin EXTHST to high level, a host device can access some of the memory areas of DM270. When in external CPU mode, ARM\_A22-ARM\_A0, EM\_CE0, EM\_WE and EM\_OE become input pins. The data bus is fixed at 16-bits and uses ARM\_D15-ARM\_D0. Access from a host device can be performed dynamically by enabling EM\_WAIT output for the host device to monitor the status of the DM270. DM270 can interrupt the host device through GIO25, after GIO25 has been configured to function as the MIRQ (interrupt) signal.

#### 4.1.2 Memory Space

The memory areas that can be accessed from an external host controller are ARM internal memory (32 Kbytes), SDRAM (16,128Kbytes), and DSP internal memory (128 Kbytes). However, the memory map addresses differ from the usual ARM memory map. The following Table shows the memory map of DM270 as seen from the external CPU.

| Memory Area         | Start address | End address | Size (byte) |
|---------------------|---------------|-------------|-------------|
| EHIF registers      | 0x00:0000     | 0x00:007F   | 128         |
| ARM Internal Memory | 0x00:8000     | 0x00:FFFF   | 32 K        |
| DSP Internal Memory | 0x02:0000     | 0x03:FFFF   | 128 K       |
| SDRAM               | 0x04:0000     | 0xFF:FFFF   | 16,128 K    |

#### 4.1.3 Specifications

- ❑ The data bus is fixed at 16-bits and uses ARM\_D15-ARM\_D0. Both read and write are half-word (16 bit) access.
- ❑ Access from a host device can be performed dynamically by enabling EM\_WAIT output for the host device to monitor the status of the DM270 ('0':Busy, '1':ready). WAITEN register of the External CPU Interface must be set in advance to enable the EM\_WAIT output.
- ❑ The access time seen from the host device varies depending on ARM, DSP and SDRAM clock frequency settings. (Refer to 4.4 Access Time)
- ❑ 32byte Burst/Continuous Burst access to SDRAM is supported via sixteen 16bit data registers.
- ❑ Host device can issue an interrupt signal to the ARM by setting the register INT.
- ❑ If a read is performed continuously after read, EM\_OE or EM\_CE0 must once return to high level. RTRG register must be set in case EM\_OE becomes high level and EM\_CE0 stays low.

#### 4.1.4 Access Time

The External Host I/F module operates by the ARM clock of the DM270. The access time seen from the host device varies depending on ARM, DSP and SDRAM clock frequency settings. The number of cycles of the ARM clock is shown below.

| Access Type                                | Time                    | Notes                                  |
|--------------------------------------------|-------------------------|----------------------------------------|
| EHIF register/SDRAM burst access/AIM write | 2 + $\alpha$ cycles     | ( $\alpha$ :Input and/or Output delay) |
| AIM read                                   | 3 + $\alpha$ cycles     |                                        |
| SDRAM single access/DSP memory             | 3 + n + $\alpha$ cycles | (n:Peripheral wait cycles)             |

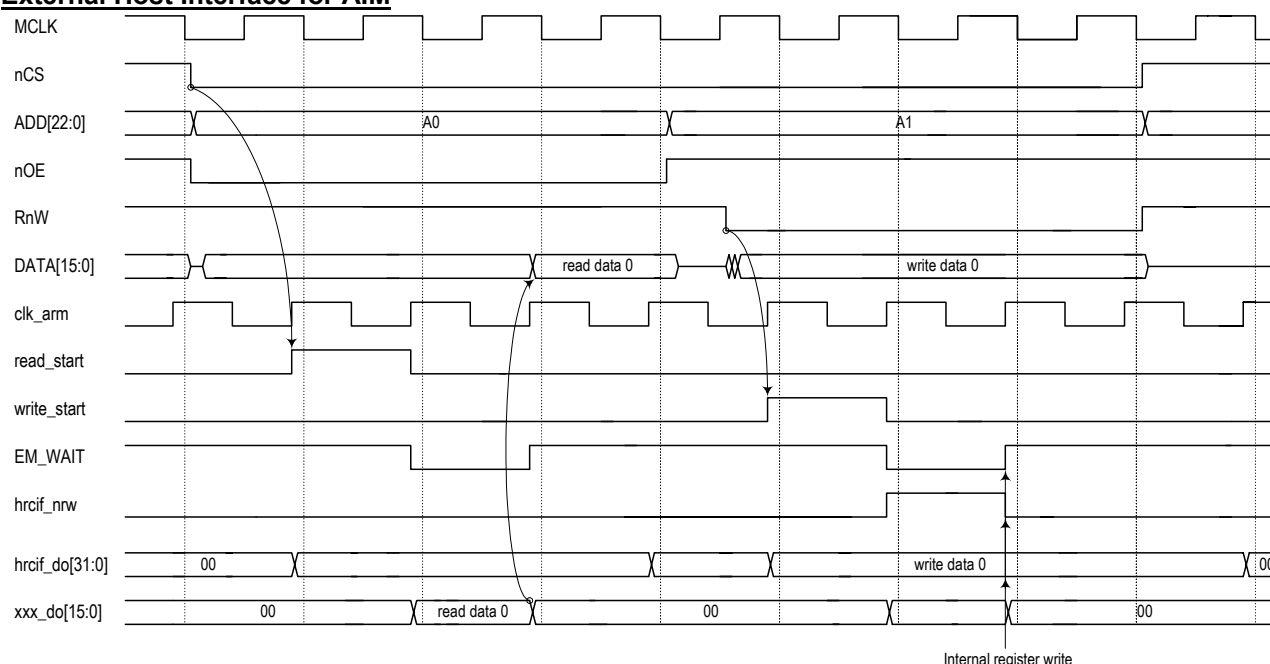
### 4.1.5 SDRAM Access

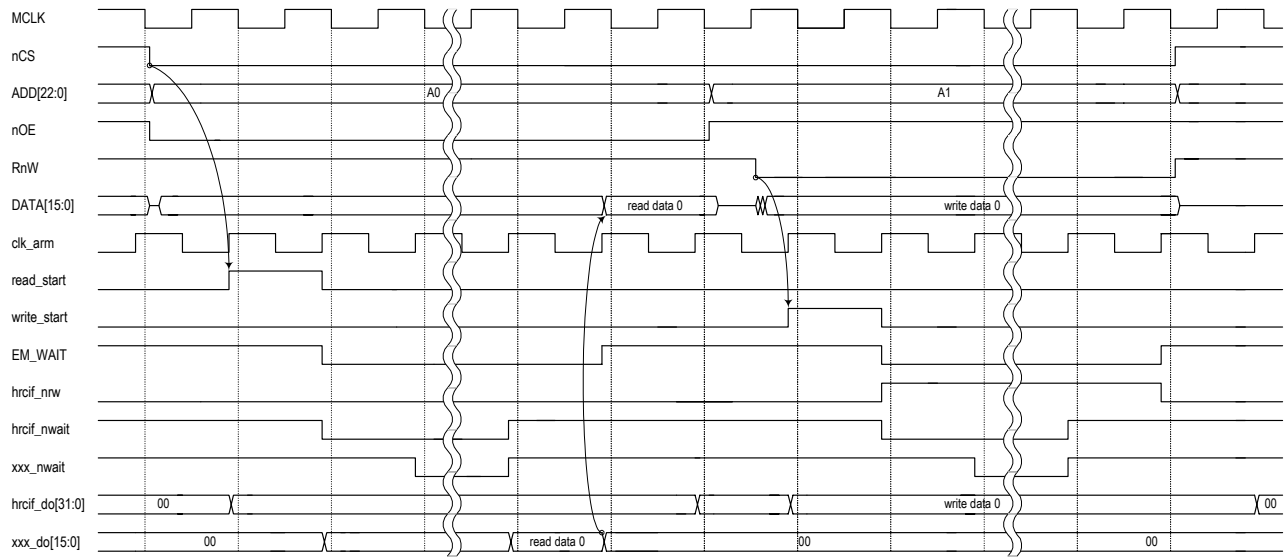
Single/32byte Burst/Continuous Burst access to SDRAM is supported. Sixteen 16bit data registers are used for Burst/Continuous Burst access. When in Continuous Burst transfer mode, data of the specified length in 32 byte units are transferred to and from SDRAM. There are 2 modes for Continuous Burst access.

- **Mode1:** Address decode enabled. Register, AIM and HPIB accesses are allowed during Continuous Burst access. There are some registers to monitor the status of EHIF during Continuous Burst access which are:
  1. **Burst transfer wait flag:** To check the Burst transfer status every 16 accesses. This is for a host device which does not have a WAIT input.
  2. **SDRAM access wait flag:** To check the access status between EHIF and SDRC before you start access to AIM/HPIB during Continuous Burst transfer, or before you end the Continuous Burst transfer.
- **Mode2:** No address decode. No other access is allowed during DMA.

### 4.1.6 Timing

#### External Host Interface for AIM



**External Host Interface for hand shake peripherals (HPIB)**

## 5 AF/AE/AWB Statistics Engine

DM270 provides real-time **3A (AE/AWB/AF)** statistics generation function in hardware. 3A statistics function can be performed either before WB or after gamma conversion. When 3A statistics calculation is completed, an interrupt will be generated to inform ARM of its completion. ARM then uses the statistics to compute the optimal 3A control parameters.

AE/AWB statistics are calculated using 192 windows. The windows are the same size and organized as an m-by-n matrix. The organization of the matrix, the size of the windows and the method of calculation are user-selectable. AF statistics is calculated by up to 36 windows. The position and the size of each window can be independently set by software.

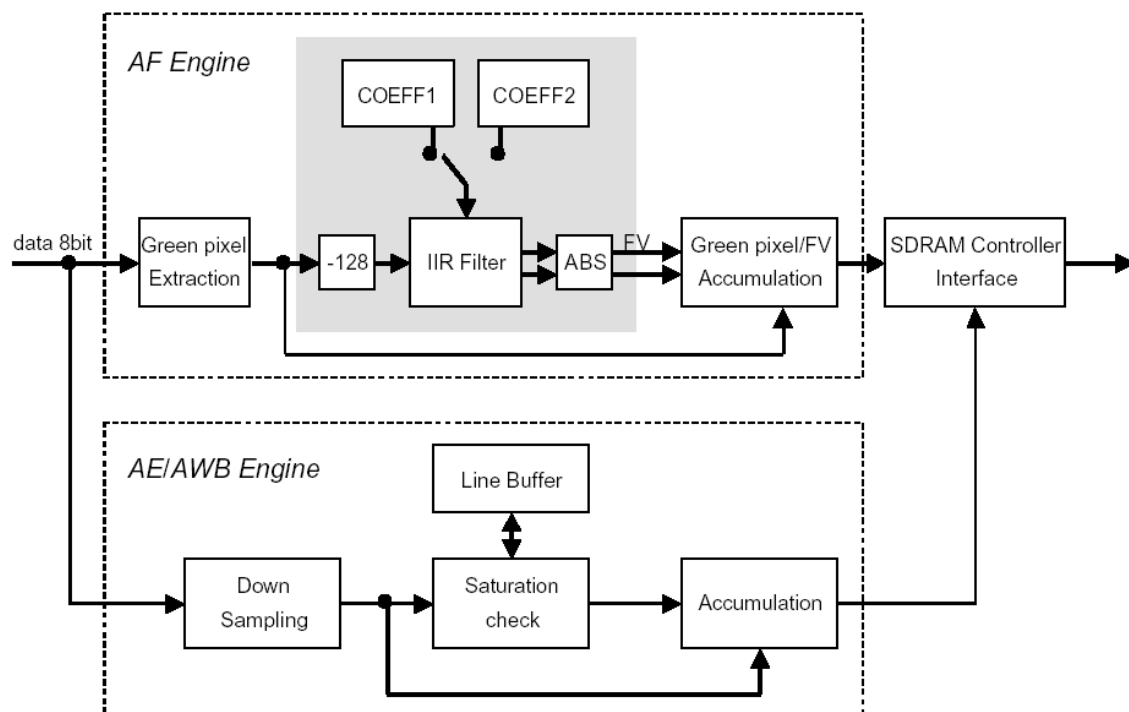
### 5.1 Features

#### 5.1.1 Auto Exposure, Auto White Balance

- ❑ Total number of windows is 192.
- ❑ The base position for AE/AWB statistics window matrix can be configured by the user
- ❑ Number of windows in horizontal/vertical direction is from 1 to 12(H),8(V)
- ❑ Total number of pixels in horizontal/vertical direction is from 2 to 256(H), 64(V)
- ❑ Sampling step for calculation in horizontal/vertical direction is from 2 to 32. Total number of 2x2 blocks in a window is up to 256.
- ❑ Outputs for each window are the accumulated R, Gr, Gb and B. The results are 16-bit integer stored in SDRAM readable by ARM

#### 5.1.2 Auto Focus

- ❑ Total number of windows is up to 36
- ❑ The calculation can be performed on Green pixels only data in the window.
- ❑ The position, size of each window can be independently set by software.
- ❑ Two biquad filters are used to filter the green or Y pixel data. The coefficients of both filters are 10-bit wide and can be set by software.



**Block Diagram of 3A Hardware Engine**

#### ***AF Engine***

- Green pixel accumulation for each Poxel.
- Focus Value (FV) accumulation for each Poxel.

#### **[Peak mode]**

- Accumulate the maximum FV of each line in a Poxel.
- Accumulate the peak position

#### **[Sum mode]**

- Accumulate FV in a Poxel.

#### ***AE/AWB Engine***

- Accumulate the pixels in the sub-sampled 2x2 blocks for each Window (AVE1).
- Accumulate the pixels which are clipped to the limit value in the sub-sampled 2x2 blocks (AVE2).
- Count the number of the sub-sampled 2x2 blocks which are not saturating (CNT2).



## 6 I<sup>2</sup>C Interface

The DM270 includes an Inter Integrated Circuit (I<sup>2</sup>C) serial interface that can act as single master. Both 100kbps and 400kbps I<sup>2</sup>C-bus system is implemented. The interface uses 7bit/10bit addressing. The DM270 initiates and terminates transfers and generates clock signals.

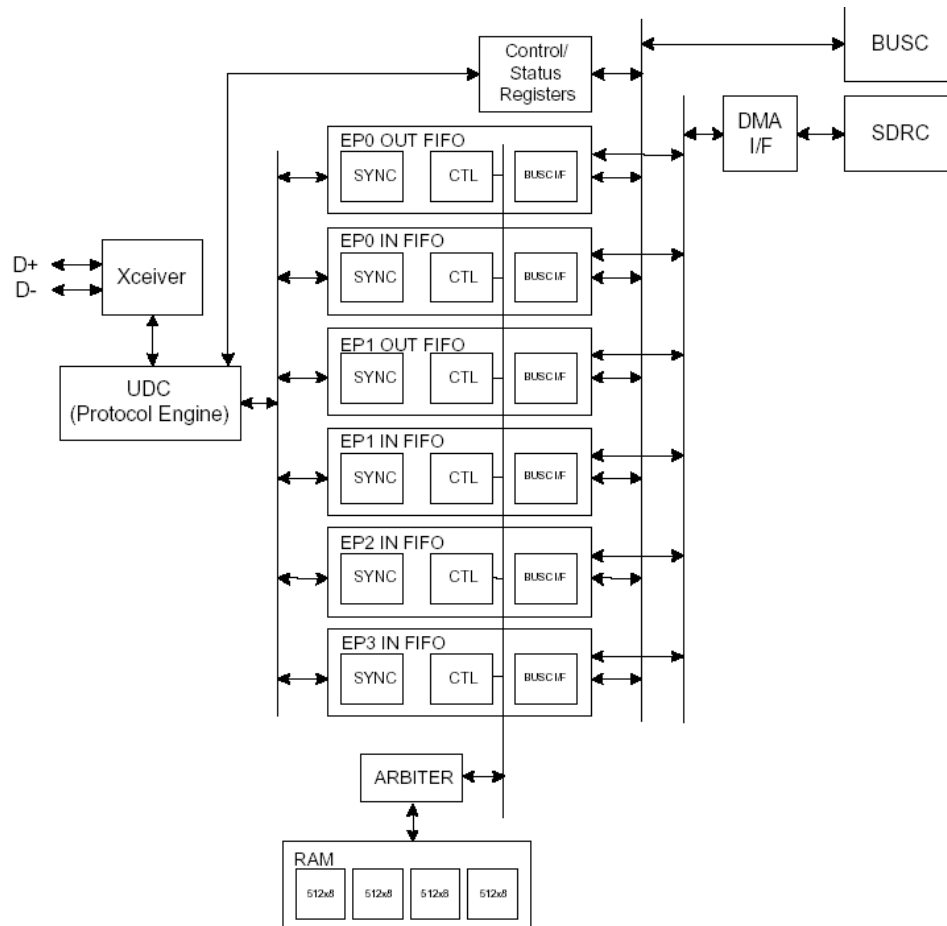
## 7 ARM Instruction cache (I-cache)

I-cache in DM270 contains 8kbyte RAM for caching instruction. I-cache selects only instruction mapped external memory (0x100000-0x80FFFF) . Below are I-cache features.

- ❑ 4-word (128bit) line size
- ❑ Total 512-line (256-line x 2way)
- ❑ Update data by LRU(Least Recently Used) algorithm
- ❑ Check tag-RAM according to address from ARM and execute below action in one wait cycle.
- ❑ When HIT, ARM reads the data stored in RAM in I-cache
- ❑ When MISS, external memory access starts
- ❑ One valid bit per 2byte data in tag-RAM
- ❑ No Dirty bit in tag-RAM (no support write)
- ❑ Support I-cache disable(bypass) mode
- ❑ Selectable 1word/4word access (4word burst is supported only SDRAM access)
- ❑ Support cache clear function

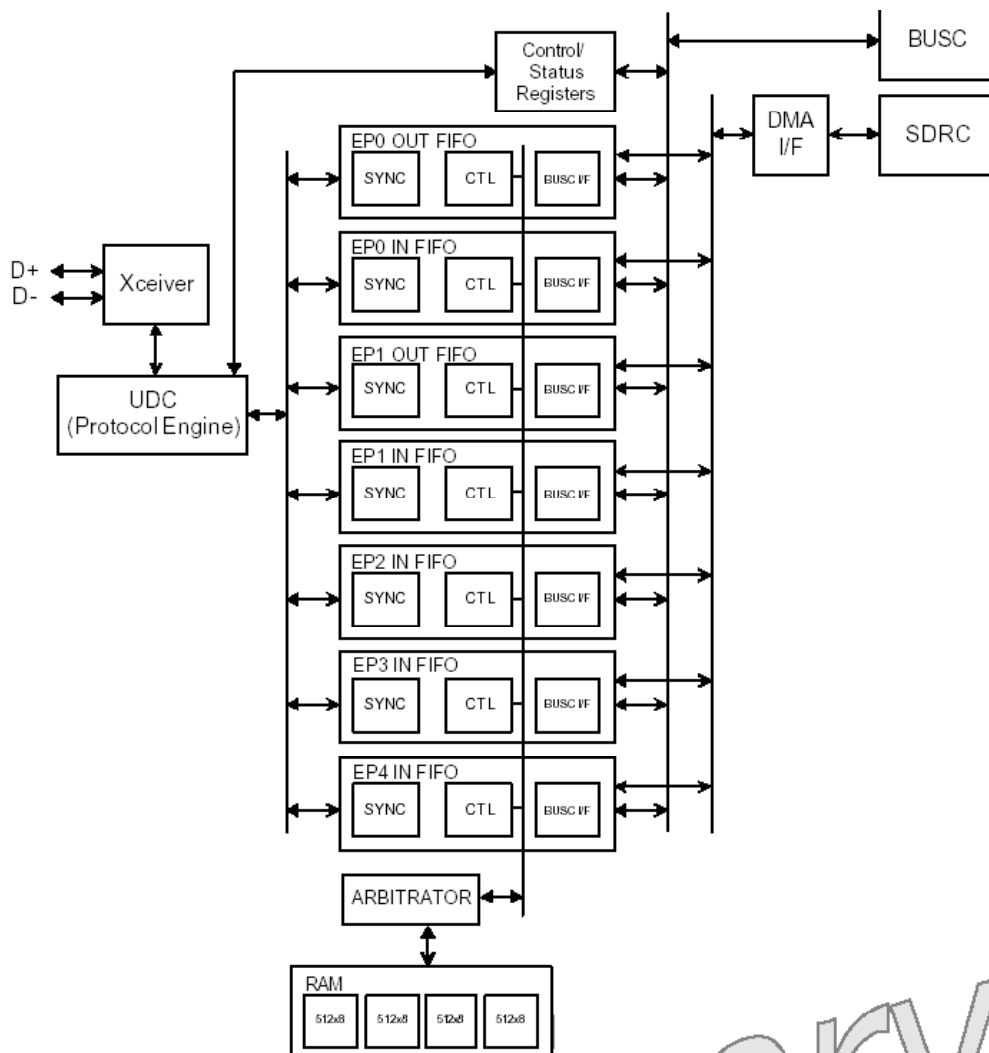
# 8 USB

USB module in DSC25 has a total of six FIFOs, each associated with an Endpoint. The structure of each FIFO is the same except for the direction and the buffer size. The maximum buffer size is 256 bytes for EP0 and EP2 while it is 2048 bytes for EP1 and EP3.



**DSC25: Block Diagram of USB**

In DM270, USB module has a built-in xceiver and 2K bytes RAM. USB module has 7 FIFOs and each FIFO is associated each Endpoint. The structure of each FIFO is same except for the direction and the buffer size. The maximum buffer size is 256 bytes for EP0 and EP2 while it is 2048 bytes for EP1, EP3 and EP4.



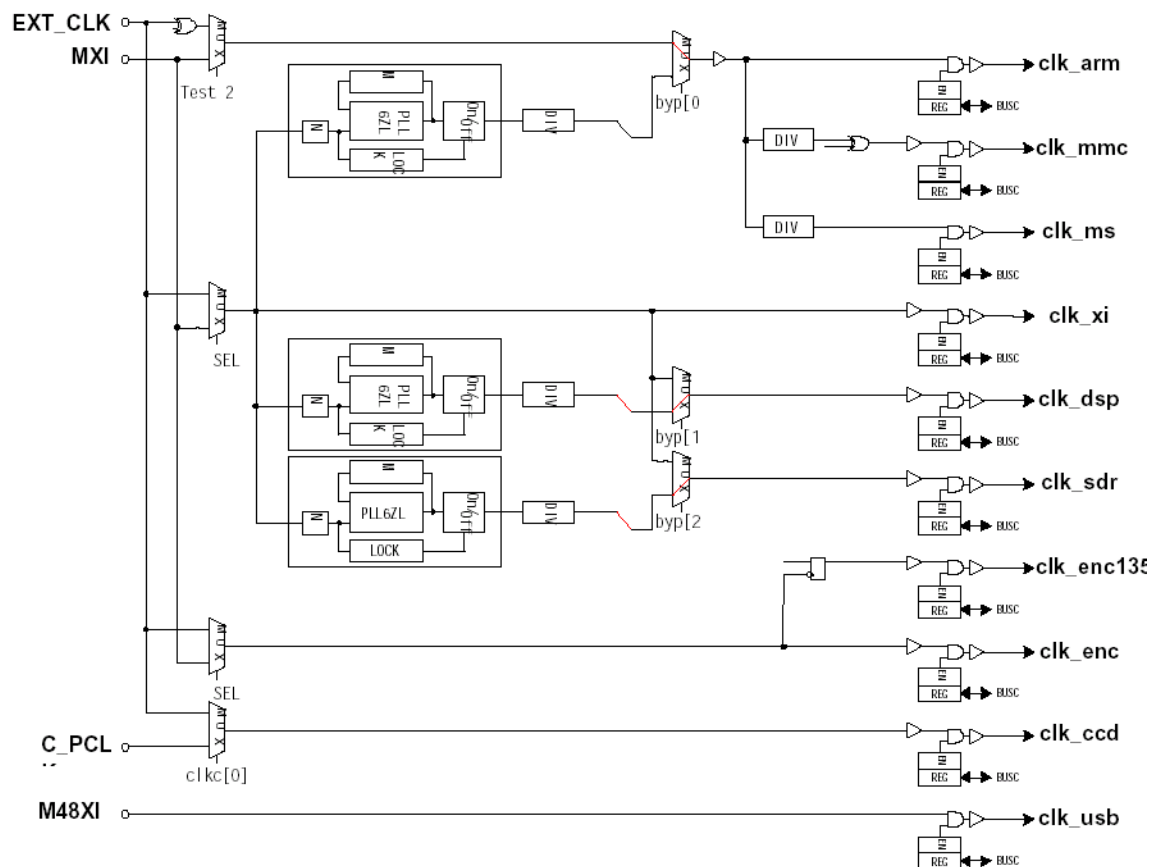
**DM270: Block Diagram of USB**

## 9 PLL / Clock Controller

In DSC25, the CLKC has three analog PLLs built into it, and you can freely set the frequencies of the ARM clock, SDRAM clock, and DSP clock using these PLLs.

The Clock Controller generates the following 10 signals:

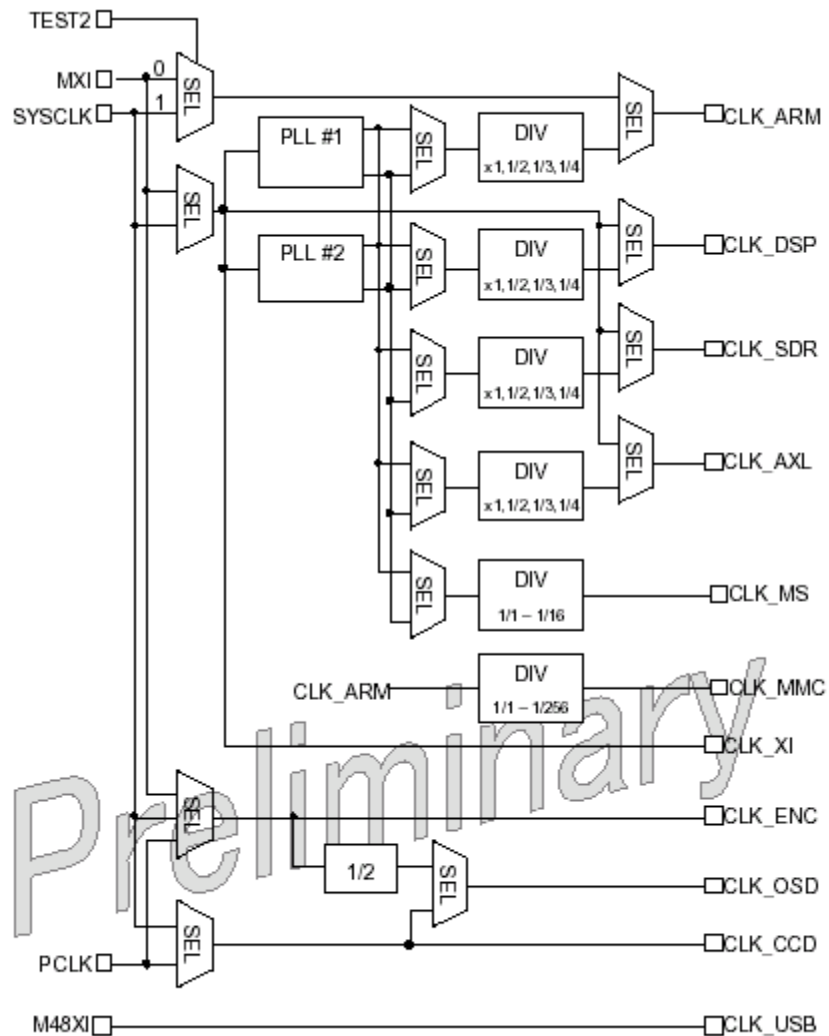
- |                      |                                                      |
|----------------------|------------------------------------------------------|
| 1. <b>CLK_ARM</b>    | ARM clock for ARM core and its related modules       |
| 2. <b>CLK_DSP</b> :  | DSP clock for DSP core and its related modules       |
| 3. <b>CLK_SDR</b> :  | SDR clock for external SDRAM and its related modules |
| 4. <b>CLK_XI</b>     | X'tal clock such as TMR, WDT                         |
| 5. <b>CLK_CCD</b>    | Video interface clock                                |
| 6. <b>CLK_ENC135</b> | OSD clock                                            |
| 7. <b>CLK_ENC</b>    | Video encoder clock                                  |
| 8. <b>CLK_USB</b>    | USB clock                                            |
| 9. <b>CLK_MMC</b>    | Multimedia Card / Secure Digital clock               |
| 10. <b>CLK_MS</b>    | Memory Stick clock                                   |



**DSC25 Clock Controller Block**

In DM270, CLKC has two analog PLLs built into it. It can generate an additional signal:

**CLK\_AXL** Accelerator clock for iMX,VLCD and Image Buffers



**DM270 Clock Controller Block**

## 10 General I/O

The DM270 has 34 General-Purpose I/O (GIO) ports compared to the DSC25's 32 GIO ports. In addition, the DM270 has added additional features to the GIO capability

- ❑ A total of 16 GIOs can be configured as external interrupts (vs. 8 in DSC25)
- ❑ Each external interrupts can be configured to trigger on rising edge, falling edge, or either edge
- ❑ DM270 adds ability to configure GIO signals for immunity to chatter

# 11 Interrupt Controller

DSC25 Interrupt Controller (INTC) has 28 interrupt channels: 20 peripheral interrupt channels and 8 external interrupt channels. In DM270, the Interrupt Controller (INTC) has 39 interrupt channels: 23 internal peripheral interrupt channels and 16 external interrupt channels. Table below lists each interrupt and the interrupt's corresponding function.

| Interrupt | Interrupt Source     | Description                                                                                       |
|-----------|----------------------|---------------------------------------------------------------------------------------------------|
| INT38     | CLKC                 | WAKE UP                                                                                           |
| INT37     | I <sup>2</sup> C     | I <sup>2</sup> C                                                                                  |
| INT36     | WDT                  | WDT                                                                                               |
| INT35     | PREVIEW              | Occurs at the end of frame(configured by HSIZE/VSIZ registers) when SDRAM input mode is selected. |
| INT34     | External interrupt15 | IRP15 bit IRQPORT register GIO15                                                                  |
| INT33     | External interrupt14 | IRP14 bit IRQPORT register GIO14                                                                  |
| INT32     | External interrupt13 | IRP13 bit IRQPORT register GIO13                                                                  |
| INT31     | External interrupt12 | IRP12 bit IRQPORT register GIO12                                                                  |
| INT30     | External interrupt11 | IRP11 bit IRQPORT register GIO11                                                                  |
| INT29     | External interrupt10 | IRP10 bit IRQPORT register GIO10                                                                  |
| INT28     | External interrupt9  | IRP9 bit IRQPORT register GIO9                                                                    |
| INT27     | External interrupt8  | IRP8 bit IRQPORT register GIO8                                                                    |
| INT26     | External interrupt7  | IRP7 bit IRQPORT register GIO7                                                                    |
| INT25     | External interrupt6  | IRP6 bit IRQPORT register GIO6                                                                    |
| INT24     | External interrupt5  | IRP5 bit IRQPORT register GIO5                                                                    |
| INT23     | External interrupt4  | IRP4 bit IRQPORT register GIO4                                                                    |
| INT22     | External interrupt3  | IRP3 bit IRQPORT register GIO3                                                                    |
| INT21     | External interrupt2  | IRP2 bit IRQPORT register GIO2                                                                    |
| INT20     | External interrupt1  | IRP1 bit IRQPORT register GIO1                                                                    |
| INT19     | External interrupt0  | IRP0 bit IRQPORT register GIO0                                                                    |
| INT18     | MMC/SD MS            | MMC/SD Interface and MS Interface                                                                 |
| INT17     | MMC/SD MS            | MMC/SD Interface and MS Interface                                                                 |
| INT16     | EMIF 1               | BUSINTEN register                                                                                 |
| INT15     | EMIF 0               | At the end of DMA transfer DMACTL register                                                        |
| INT14     | USB                  | USB                                                                                               |
| INT13     | UART1                | UART1                                                                                             |
| INT12     | UART0                | UART0                                                                                             |
| INT11     | DSP                  | DSP_HINT register (DSP)                                                                           |
| INT10     | External CPU I/F     | External CPU I/F REGISTER                                                                         |
| INT9      | Serial I/F 1         | At the end of DMA transfer                                                                        |
| INT8      | Serial I/F 0         | At the end of DMA transfer                                                                        |
| INT7      | OSD                  | At the falling edge of VSYNC                                                                      |
| INT6      | CCD Controller       | At the falling edge of C_WEN port                                                                 |
| INT5      | CCD Controller       | VDINT1 register (CCD Controller)                                                                  |
| INT4      | CCD Controller       | VDINT0 register (CCD Controller)                                                                  |
| INT3      | TIMER3               | TIMER3                                                                                            |
| INT2      | TIMER2               | TIMER2                                                                                            |
| INT1      | TIMER1               | TIMER1                                                                                            |
| INT0      | TIMER0               | TIMER0                                                                                            |



# 12 SDRAM controller

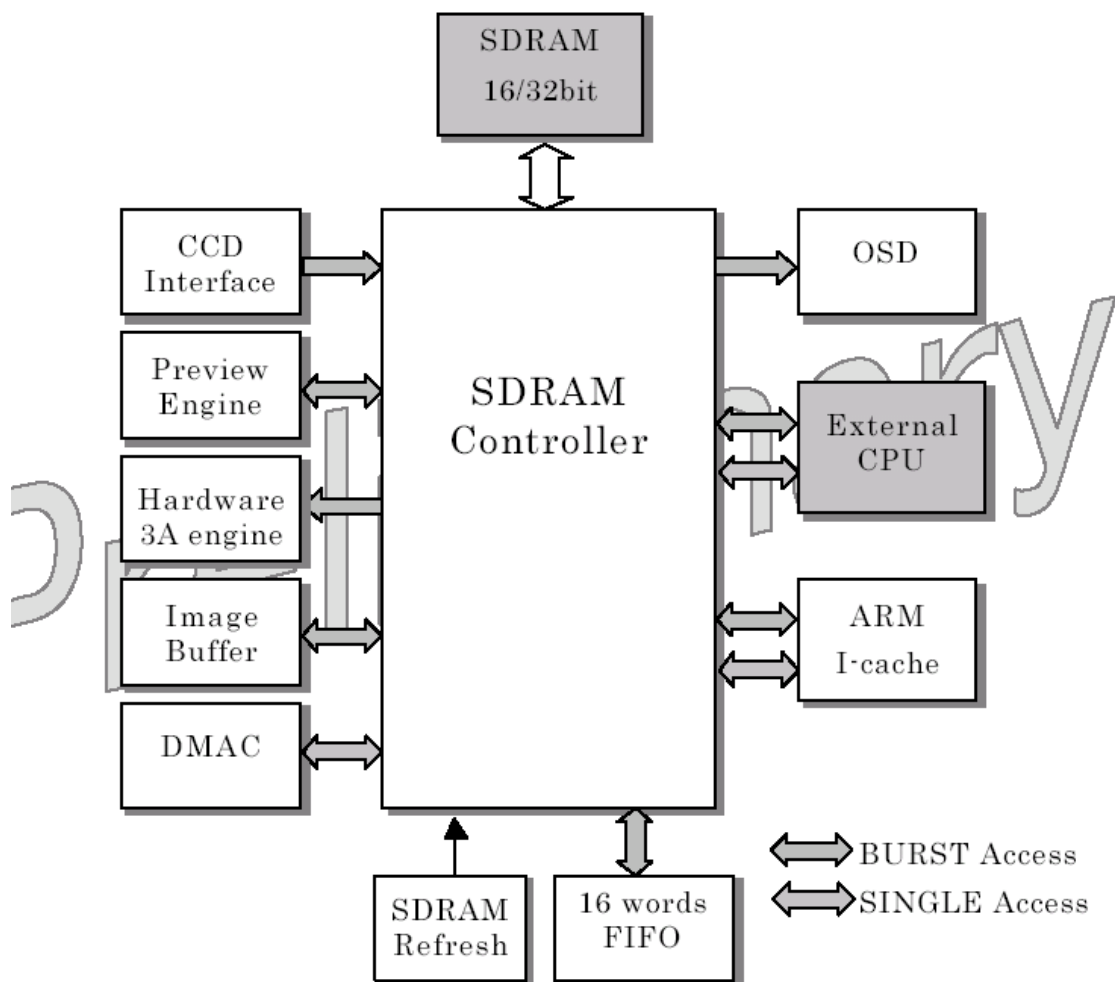
The differences between SDRAM controller in DSC25 and DM270 are listed below:

## DSC25:

- Supports 16/64/128/256M SDRAM
- CAS latency 1/2/3 can be selected

## DM270:

- Supports 16/64/128/256/512M SDRAM
- CAS latency 2/3 can be selected



**DM270 SDRAM Controller Block**

# 13 DSP Subsystem

## 13.1 Memory map

### DSC25:

The DSP subsystem offers 32 kwords (1 word = 16 bits) of DSP internal DARAM memory, 16 kwords of ROM memory, and 16 kwords of shared memory. The program space and the data memory spaces span the addresses 0x0h to 0xFFFFh.

### DM270:

The DSP subsystem offers 32 K words (1 word = 16 bits) of DSP internal DARAM memory, 32 K words of DSP internal SARAM memory, 32 K words of ROM memory, and 16 K words of shared RAM memory. The program space and the data memory spaces span the addresses 0x0h to 0x1FFFFh. Figure given below shows the DM270 DSP memory map after reset.

|      | DATA                                                   | PROGRAM<br>(PAGE0)                           | PROGRAM<br>(PAGE1)              |
|------|--------------------------------------------------------|----------------------------------------------|---------------------------------|
| 0000 | MMR                                                    |                                              |                                 |
| 007F | Scratched-Pad                                          | Reserved                                     | Reserved                        |
| 0080 | Data/Program<br>RAM<br>(OVLY=1)                        | Data/Program<br>RAM<br>(OVLY=1)              | Data/Program<br>RAM<br>(OVLY=1) |
| 7FFF |                                                        |                                              |                                 |
| 8000 | Data<br>RAM                                            | Program<br>ROM                               | Reserved                        |
| BFFF |                                                        |                                              |                                 |
| C000 | ROM<br>(DROM=1)<br>or<br>Image Buffer<br>(DROM=0)      | ROM<br>(MP/MC=0)                             |                                 |
| FFFF |                                                        |                                              | Program<br>RAM                  |
| FF00 | Reserved<br>(DROM=1)<br>or<br>Image Buffer<br>(DROM=0) | Interrupts<br>& Reserved<br>ROM<br>(MP/PC=0) |                                 |
| FFFF |                                                        |                                              |                                 |

## 13.2 I/O memory space

The C54x devices offer an I/O-memory space in addition to the program memory and data-memory spaces. The I/O-memory space is a 64K-word address space (0000h–FFFFh) and exists only external to the DSP device. Image Buffer registers, iMX registers, and VLCD registers are mapped to I/O space. Signals to control the Image buffer clock, iMX buffer clock, and VLCD buffer clock are mapped to an I/O space. A signal to interrupt the ARM from the DSP is also mapped to an I/O space. Tables below show the address allocation for DSC25 and DM270.

| I/O address     | Module name          |
|-----------------|----------------------|
| 0x0000h-0x0009h | Image Buffer         |
| 0x0010h-0x0014h | iMX                  |
| 0x0020h-0x005Dh | VLCD                 |
| 0xFFFFeh        | CLOCK Control        |
| 0xFFFFh         | DSP to ARM Interrupt |

**DSC25 I/O Space**

| I/O address     | Module name          |
|-----------------|----------------------|
| 0x0000h-0x0009h | Image Buffer         |
| 0x0010h-0x0017h | iMX                  |
| 0x0020h-0x005Fh | VLCD                 |
| 0x0080h-0x0087h | HPIB                 |
| 0xFFFFeh        | CLOCK Control        |
| 0xFFFFh         | DSP to ARM Interrupt |

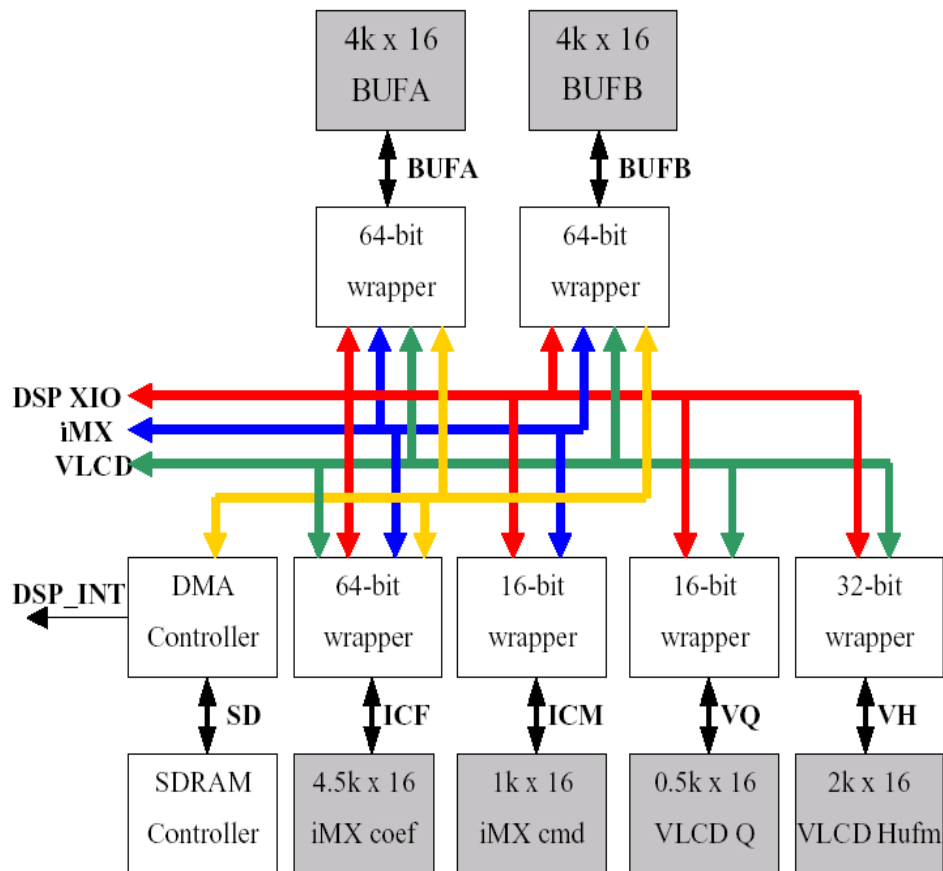
**DM270 I/O Space**

### 13.3 Coprocessor Subsystem

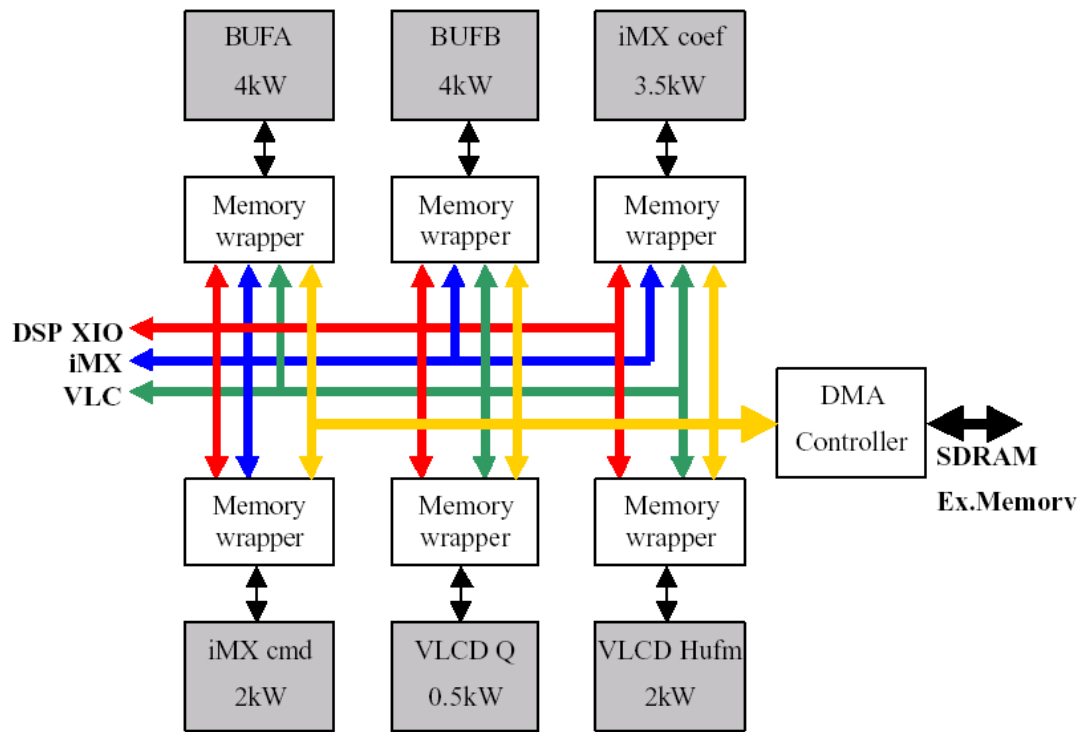
The coprocessor subsystem is part of the DSP subsystem. It is designed to enhance DSP's capability in image processing and in other computation-intensive applications.

#### 13.3.1 Block Diagram

Coprocessor subsystems for DSC25 and DM270 are shown below.



**DSC25 Coprocessor subsystem**



DM270 Coprocessor subsystem

### 13.3.2 Memory Structure

| Name             | Start Address | End Address | Size (Words) |
|------------------|---------------|-------------|--------------|
| Image Buffer A   | 0xC000        | 0xCFFF      | 4K           |
| Image Buffer B   | 0xD000        | 0xEDFF      | 4K           |
| iMX Coef. Buffer | 0xE000        | 0xEDFF      | 3.5K         |
| iMX Cmd Buffer   | 0xEE00        | 0xF5FF      | 2K           |
| VLC Q buffer     | 0xF600        | 0xF7FF      | 512          |
| VLC Huffman Buf  | 0xF800        | 0xFFFF      | 2K           |

### 13.4 Imaging Extension (iMX)

DM270 iMX is a 4-MAC version based on DSC25 iMX, with the following enhancements:

#### 13.4.1 Logical AND/OR/XOR:

Supports extracting lower bits, for post-lookup interpolation.

#### 13.4.2 MIN-ID and MAX-ID:

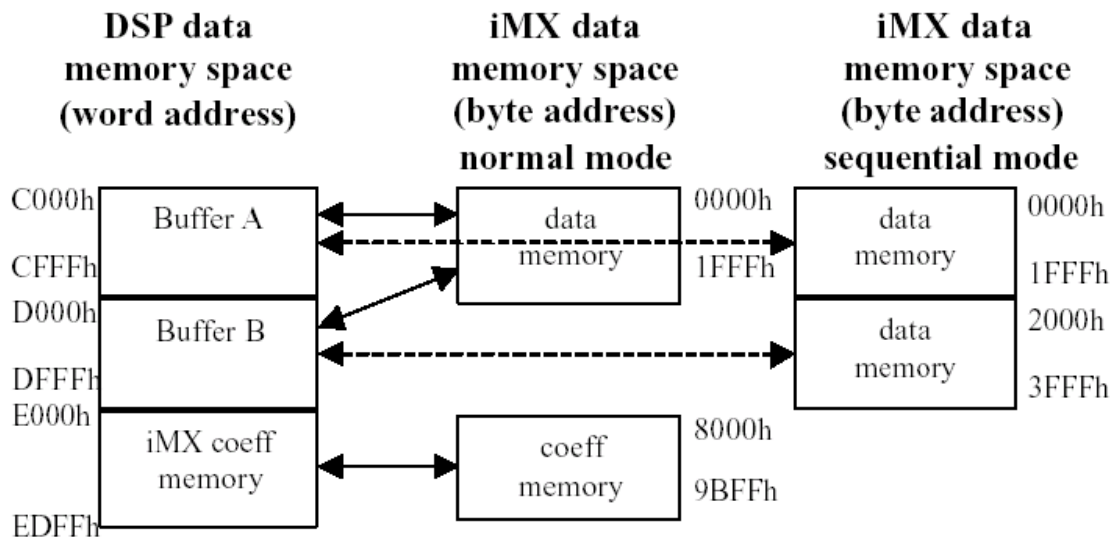
Basically preserves the pointer value to which the minimal value resides. This helps motion estimation.

#### 13.4.3 New “CALL” command

Enables re-use of iMX commands

#### 13.4.4 Sequential access to image buffer A and B for non concurrent process

DM270 has another memory interface to shared memory. It enables a sequential memory access to buffer A and B. In this mode iMX can access to image buffer A and B directly. Also DSP has to switch both buffer A and B to iMX access.

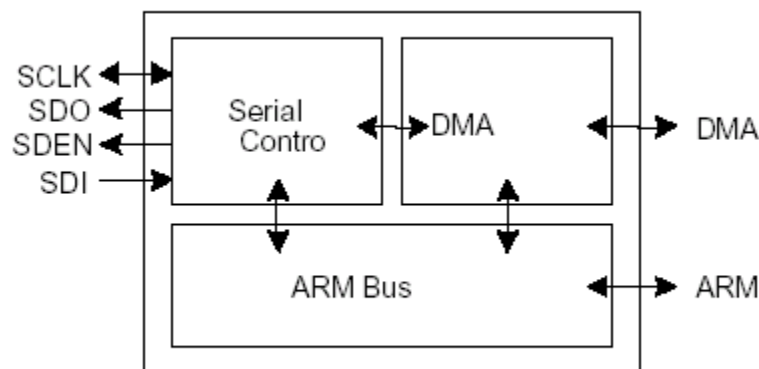


Shared Memory Between iMX and DSP

## 14 Serial Port

Both DSC25 and DM270 have a two-channel clock synchronized serial port interface (SPI). The transfer unit is 8 bits each time, and it is a simple three-line serial interface consisting of a clock line, output data line, and input data line. Serial port 0 has a DMA transfer function with SDRAM for transfer of large quantities of data. **DM270 SPI has master mode and slave mode and which is selectable. The SPI can supply the serial clock when it is used in a master mode. The SPI also can be synchronized by the external serial clock input when it is used in a slave mode.**

Both of the serial ports and the DMA channel are controlled by ARM registers.



**DM270 Serial Port Interface**

# 15 Bus Controller

The main decoding ranges are shown below

| Area Name        | Start address | End address | Size    | Bus width      | Number of wait states |
|------------------|---------------|-------------|---------|----------------|-----------------------|
| Reset vector ROM | 0x0000:0000   | 0x0000:0003 | 4 bytes | 32 bits        | 0                     |
| ARM Internal RAM | 0x0000:0004   | 0x0000:7FFF | ~32 KB  | 8, 16, 32 bits | 0                     |
| ARM Peripheral   | 0x0003:0000   | 0x0003:0FFF | 4 KB    | 16 bits        | 0†                    |
| DSP Memory       | 0x0004:0000   | 0x0004:FFFF | 64 KB   | 16 bits        | 1 minimum             |
| DAEM             | 0x0010:0000   | 0x080F:FFFF | 128 MB  | 8, 16, 32 bits | 1 minimum             |

† The number of cycles for USB register access varies depending on the internal state, and several cycles are required.

## DSC25 ARM Memory Space

| Area Name                        | Start address | End address | Size in bytes |
|----------------------------------|---------------|-------------|---------------|
| Reset vector ROM <sup>(†1)</sup> | 0x0000:0000   | 0x0000:0003 | 4             |
| ARM Internal RAM                 | 0x0000:0004   | 0x0000:7FFF | 32 K          |
| ARM Internal ROM                 | 0x0000:8000   | 0x0000:BFFF | 16 K          |
| Internal Peripheral              | 0x0003:0000   | 0x0003:0FFF | 4 K           |
| DSP Memory                       | 0x0004:0000   | 0x0005:FFFF | 128 K         |
| External Memory                  | 0x0010:0000   | 0x080F:FFFF | 128 M         |

## DM270 ARM Memory Space



# 16 ARM Boot-up Sequence

DM270 provides three different ARM boot-up sequences. There are **self boot-up mode**, **serial boot-up mode** and **HOST boot-up mode**. DSC25 provides only self boot-up mode.

## 16.1 Self boot-up mode

Upon power up and reset, the ARM program counter is reset to 0x000, which resides at the start of ARM Internal RAM. The ARM reset vector is hard coded in ROM at this location with an instruction to branch to 0x0010:0000 (start of the CS0 address space). The ARM should then execute a boot sequence from CS0 region to start the DM270 system. After turn-off the power on reset signal to first access to CS0(FLASH\_CE) region takes about 10 ARM clock cycles. In this mode, the external pin of BTSEL should be tied to low(gnd) level.

## 16.2 Serial boot-up mode

DM270 is downloadable externally through serial interface (SIF0) and UART (UART0) pins. During reset, the external CPU can access to ARM internal memory by using SIF/UART. This interface consists of four signals serial clock input (SCLK0), enable signal input (SDEN0), serial data input (RXD0) and serial data output(TXD0). The external CPU can write a necessary data to ARM internal memory (AIM), for instance boot code, and read out the data from AIM.

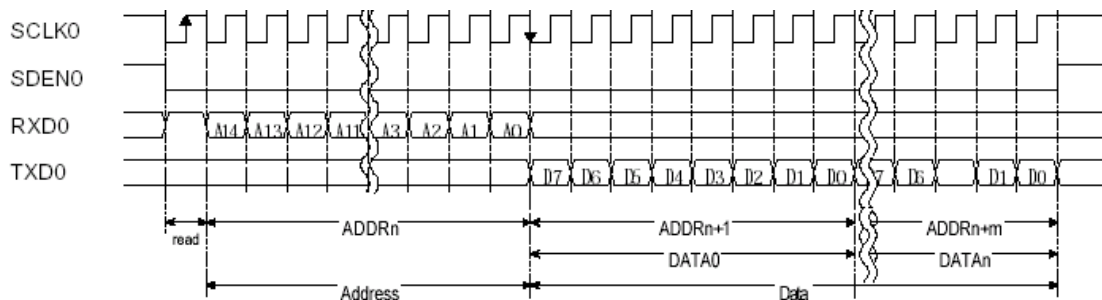
The followings show the features:

- ❑ Full addressing from 0x0 to 0x007F:FFFF accessible
- ❑ Access mode is a byte mode only
- ❑ Address can be incremented automatically after getting first address
- ❑ Support both of write and read
- ❑ Effectively during power on reset is low level, e.g. during DM270 is on reset.
- ❑ Effectively during SDEN0 is low level

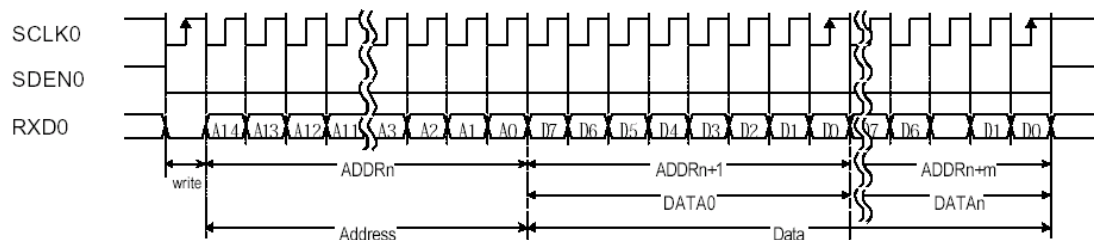
The following figures show timing diagram for serial interface. The first received bit indicates read/write command and it follows 15 bits address bits. When the command bit is low, being the write mode and when the command bit is high, being the read mode. Following the command and address bits, data of 8-bit length can be received in the write mode. In the read mode, data of 8 bits length read from AIM will be transferred through data out pin. The enable signal is

active low. The address to be accessed is incremented automatically during enable signal is low. The minimum unit of the transfer is defined 8 bits. If the data is less than 8 bits will be discarded at the last transferring unit. The interface has below protocols.

- ❑ The data to be transferred is MSB first.
- ❑ The maximum frequency of SCLK0 is 10MHz.
- ❑ Performed while the power on reset signal is in assertion.



**Serial data read timing chart**



**Serial data write timing chart**

### 16.3 External CPU boot-up mode

External CPU can access to ARM internal memory directly through EMIF bus pin. It also enables downloading an ARM boot code externally.

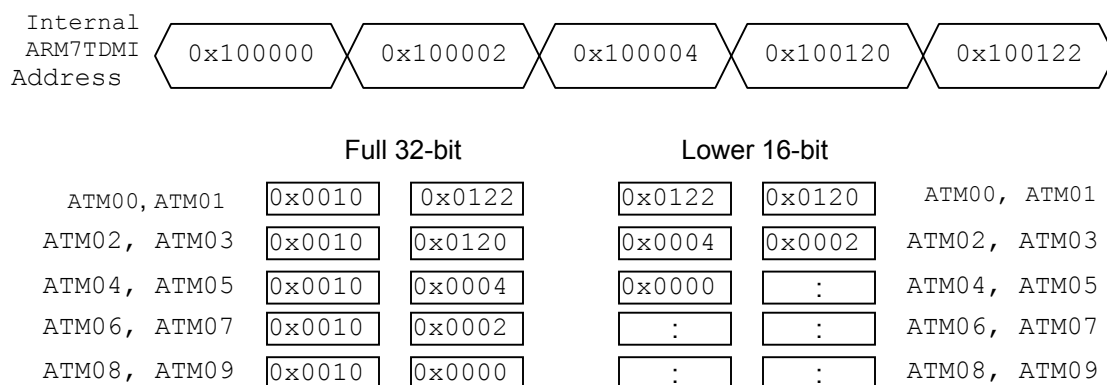
### 16.4 Hardware configuration

The BTSEL pin defines how ARM will operate after power-on reset. Table below shows pin configuration.

# 17 Address Trace Module

## 17.1 ATM Overview

Address Trace Module (ATM) provides trace function for internal ARM7TDMI. ATM has 62 sets of 16-bit registers and stores address, which internal ARM7TDMI outputs, into the registers. SW programmer can check history of internal ARM7TDMI address. Address, which is taken into ATM, is only for instruction including pre-fetch. Address for data is not taken. Buffer register size is 31 sets of full 32-bit address or 62 sets of lower 16-bit address. Stored address is last 31 or 62 sets of address. Figure - 1 shows ATM behavior and Table - 1 shows registers for ATM in TMS320DM270.



## 17.2 Notes for ATM

- LSB of address (A0) in ATM registers should be ignored since internal ARM7TDMI outputs non-meaning logic level when in thumb-mode (16-BIS).
- ATM does not support correct history after breakpoint. ATM should be locked using a\_lock bit when you view ATM register in debugger after break.
- ATM can be used regardless of ICACHE enable/disable.